BIS X Sigma Squad (IIT TP) AI Hackathon

Norm-Navigator

AI-powered BIS Standard Recommendation Engine

Problem Statement

- Indian Micro and Small Enterprises (MSEs) often spend weeks identifying applicable BIS standards
- Requires manual search across large technical documents
- Strong dependency on domain expertise
- Errors can lead to compliance risks and delays in production

There is no intelligent system to map product descriptions to BIS standards quickly and accurately.

Solution Overview

Overview

- Built a BIS Standards Recommendation Engine using hybrid RAG architecture
- Ingests official BIS dataset → converts into structured standard records
- Uses FAISS (semantic) + BM25 (keyword) for hybrid retrieval
- Returns top relevant BIS standards for natural-language compliance queries

Key Features

- Hybrid retrieval (semantic + keyword)
- Ranked top 3–5 standards
- LLM-generated grounded rationale (UI only)
- Strict dataset-bound outputs (no hallucination)

User Query → RAG Engine → Ranked BIS Standards (+ Rationale)

System Architecture

Offline Pipeline

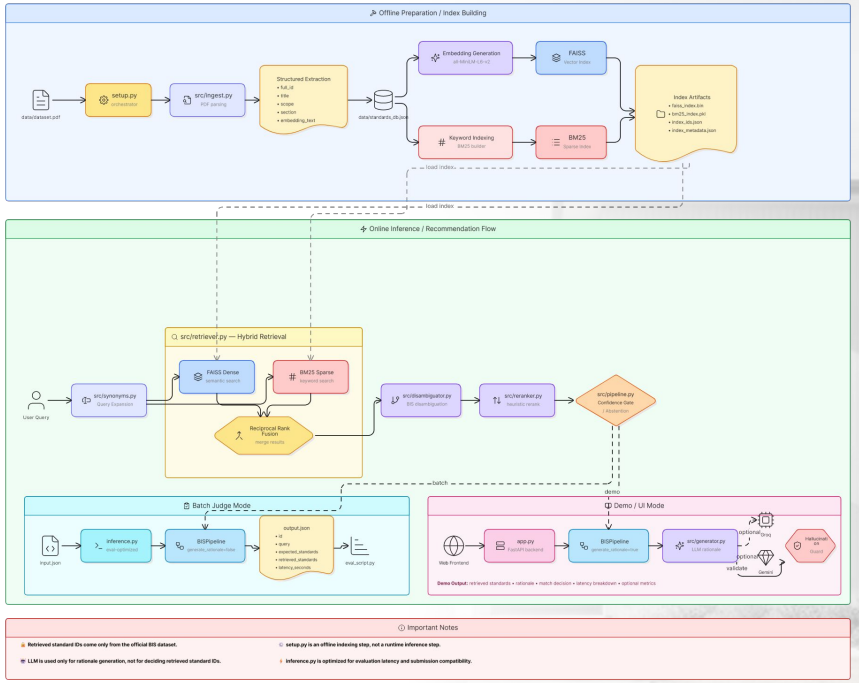
- Parse BIS PDF into standard-level records
- Generate embeddings → FAISS vector DB
- Build BM25 keyword index

Online Pipeline

- Query → synonym expansion
- Hybrid retrieval (FAISS + BM25)
- BIS-specific disambiguation
- Re-ranking
- Optional LLM rationale generation

inference.py runs in retrieval-only mode for low latency and evaluation stability

Norm-Navigator: AI-powered BIS Standard Recommendation Engine



Important Note: inference.py outputs only required fields | Web app shows rationale, latency, confidence

Chunking And Retrieval Strategy

Chunking

- One chunk = one BIS standard summary
- Extracted using “SUMMARY OF IS ...” pattern from BIS dataset
- Preserves complete semantic meaning

Retrieval

- Dense retrieval → FAISS (semantic similarity)
- Sparse retrieval → BM25 (keyword matching)
- Fusion: Reciprocal Rank Fusion (RRF)
- Semantic similarity via sentence-transformer embeddings + FAISS

Why this works

- Avoids fragmented context
- Improves ranking accuracy (MRR)
- Reduces noise & improves grounding

Future Scope

- Structured standard fingerprinting
- Material type, form & domain tracking
- Enhanced disambiguation (e.g. IS 2185)

Optimized Retrieval: Precise context for better LLM grounding

Demo Highlights

System Design Highlight:
Retrieved standards always come from indexed BIS dataset

UI shows:
Rationale
Latency breakdown
Confidence behavior

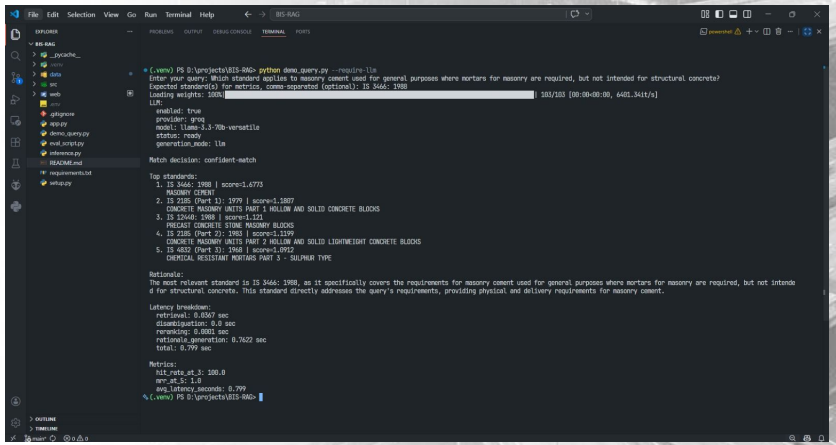
Demo Video Link

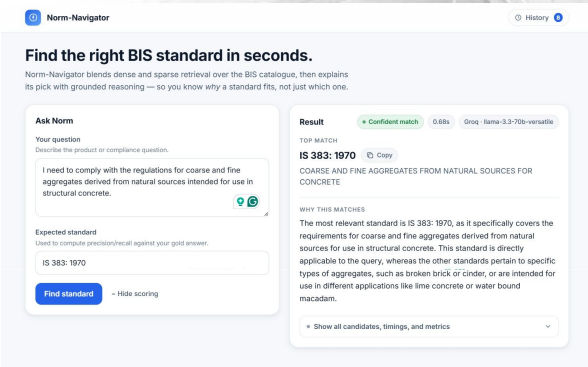
System Performance

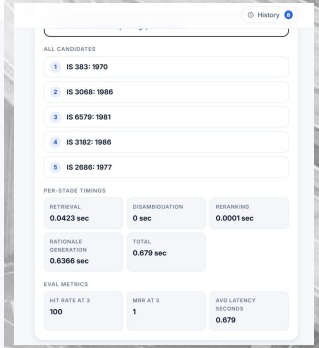
Precision Ranking
Correct standard ranked at #1

Semantic Recall
Relevant alternatives retrieved successfully

Instant Response
Average response time: < 0.3 second







Providing precise regulatory answers in real-time.

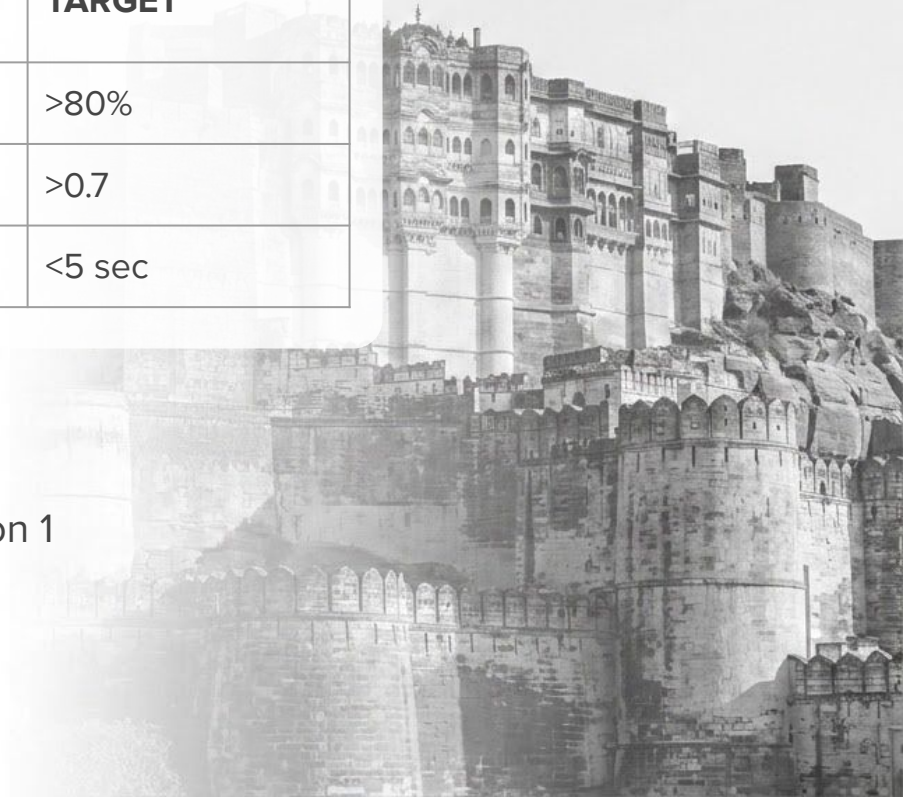
Evaluation Results

METRIC	SCORE	TARGET
Hit Rate @3	100.0	>80%
MRR @5	1.0	>0.7
Avg Latency	0.0233 sec	<5 sec

(Results of public_test_set.json)

Insights:

- Perfect MRR indicates correct ranking at position 1
- Hybrid retrieval ensures high recall + precision
- System meets latency constraints comfortably



Impact on MSEs

Key Benefits

- Reduces weeks of manual effort → **seconds**
- Eliminates need for specialized domain expertise
- Enables faster compliance and product launch
- Improves decision-making accuracy
- Scalable across multiple industries
- Reduces interpretation gap between business language and BIS terminology

Democratizing access to regulatory knowledge using AI.

Team & Acknowledgements

team thRAGg

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- **Tools:** FAISS, Sentence Transformers, BM25